Deep Forest

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Abstract

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1. Introduction

2. Preliminaries

2.1. Problem Setup

Let (X, Y) be a pair of random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where

$$X \in \mathcal{X} \subseteq \mathbb{R}^{d_x}, \qquad Y \in \mathcal{Y} = \{1, \ldots, K\}$$

denote the input feature and the class label, respectively. We consider supervised classification under the joint distribution $P_{X,Y}$.

Given a representation $T = \phi(X)$, a classifier $g : \mathcal{T} \to \mathcal{Y}$ incurs classification error

$$P_e(g, T) := \mathbb{P}\big(g(T) \neq Y\big),$$

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and the optimal error given T is

$$P_e(T) := \inf_g P_e(g, T).$$

Throughout the paper, logarithms are taken to base 2 unless otherwise stated.

2.2. Deep Forest (gcForest) Bypass Architecture

We formalize the *gcForest (bypass)* architecture as a layered representation system with a bypassed information flow. Let $X \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$ denote the original input and let $Z^{(0)}$ be an initial feature representation (e.g., $Z^{(0)} = X$ or a simple transformation of X).

Bypassed Information State.. For each layer $\ell \geq 1$, define the layer input state

$$U^{(\ell)} := (X, Z^{(\ell-1)}).$$

That is, each layer has direct access to the original input X via concatenation, together with only the *most recent* new feature $Z^{(\ell-1)}$.

Layer Update.. A depth-L gcForest induces the following bypass recursion:

$$Z^{(\ell)} = f^{(\ell)}\big(U^{(\ell)}\big) = f^{(\ell)}\big(X, Z^{(\ell-1)}\big), \qquad \ell = 1, \dots, L,$$

where $f^{(\ell)}$ is the layer-wise feature constructor implemented by a collection of forests at layer ℓ.

Forest-Based Constructor.. Let $\mathcal{H}$ be a hypothesis class of tree-ensemble predictors. At layer ℓ, a set of M_ℓ forests $\{h_m^{(\ell)}\}_{m=1}^{M_\ell} \subset \mathcal{H}$ is trained on inputs $U^{(\ell)}$ and each forest outputs a class-probability vector

$$h_m^{(\ell)} : \mathcal{X} \times \mathcal{Z} \to \Delta(\mathcal{Y}).$$

We define the newly constructed feature as the (possibly concatenated) forest outputs:

$$Z^{(\ell)} := \psi^{(\ell)}\Big(h_1^{(\ell)}(U^{(\ell)}), \dots, h_{M_\ell}^{(\ell)}(U^{(\ell)})\Big),$$

where $\psi^{(\ell)}$ denotes a deterministic aggregation operator (e.g., concatenation or averaging) that maps $\Delta(\mathcal{Y})^{M_\ell}$ into the feature space $\mathcal{Z}$.

Forwarded Quantity.. Importantly, only the *current* new feature $Z^{(\ell)}$ is forwarded to the next layer (together with X through the bypass). Equivalently, the entire depth-L model maintains the evolving state $(X, Z^{(\ell)})$ and does not accumulate all previous layer features in the forward signal.

2.3. Prediction with Restricted Model Families

For each layer ℓ, we associate a predictive model family $\mathcal{V}^{(\ell)}$ that represents the class of predictors that can be applied to $Z^{(\ell)}$. In this work, $\mathcal{V}^{(\ell)}$ corresponds to forest-based predictors with bounded complexity, reflecting computational and statistical constraints.

Given a predictive family $\mathcal{V}$ and a representation T, the optimal achievable classification error under $\mathcal{V}$ is

$$P_e^{\mathcal{V}}(T) := \inf_{g \in \mathcal{V}} \mathbb{P}\big(g(T) \neq Y\big).$$

2.4. Information-Theoretic Quantities

Definition 1 (Entropy and Mutual Information)**.** For a discrete random variable U with distribution $p(u)$, its entropy is defined as

$$H(U) := \mathbb{E}\big[-\log p(U)\big].$$

For a pair of random variables (U, V), the mutual information is

$$I(U; V) := \mathbb{E}\left[\log \frac{p(U, V)}{p(U)p(V)}\right].$$

These quantities characterize the amount of label-relevant information preserved in a representation. To relate information-theoretic measures to classification performance, we recall a classical inequality due to Fano.

Lemma 1 (Fano-Type Inequality)**.** *Let Y be a discrete random variable taking values in $\mathcal{Y}$ with $|\mathcal{Y}| = K$, and let $\widehat{Y} = g(U^{(\ell)})$ be an arbitrary (possibly randomized) estimator based on the representation $U^{(\ell)}$, such that $Y \to U^{(\ell)} \to \widehat{Y}$ forms a Markov chain. Define the probability of error*

$$P_e := \Pr(\widehat{Y} \neq Y).$$

Then,

$$H(P_e) + P_e \log(K-1) \;\geq\; H(Y \mid U^{(\ell)}),$$

and consequently,

$$P_e \;\geq\; \frac{H(Y \mid U^{(\ell)}) - 1}{\log(K-1)} \;=\; \frac{H(Y) - I(Y;U^{(\ell)}) - 1}{\log(K-1)}.$$

Since the entropy $H(Y)$ is entirely determined by the data-generating distribution and does not depend on the learned representation, it can be treated as a constant in this analysis. Lemma 1 therefore shows that any reduction in classification error necessarily requires an increase in the mutual information $I(Y;U^{(\ell)})$ between the representation and the label.

Lemma 2 (DPI and Bypass Concatenation)**.** *Assume Z is constructed from X (possibly via a randomized mapping) such that $Y \to X \to Z$ forms a Markov chain. Then the data processing inequality implies*

$$I(Y;Z) \leq I(Y;X).$$

Moreover, for the concatenated state $U := (X, Z)$, we have

$$I(Y;U) = I\big(Y;(X,Z)\big) = I(Y;X).$$

Proof. The first claim follows from the data processing inequality applied to the Markov chain $Y \to X \to Z$.

For the second claim, note first that adding a component cannot decrease mutual information:

$$I\big(Y;(X,Z)\big) \geq I(Y;X),$$

since X is a function of (X, Z).

On the other hand, by the chain rule,

$$I\big(Y;(X,Z)\big) = I(Y;X) + I(Y;Z \mid X).$$

The Markov condition $Y \to X \to Z$ is equivalent to $I(Y;Z \mid X) = 0$, hence $I(Y;(X,Z)) = I(Y;X)$. □

Lemma 2 shows that a single-layer representation $Z^{(1)} = f^{(1)}(X)$ cannot increase Shannon mutual information with the label, i.e., $I(Y;Z^{(1)}) \leq I(Y;X)$. In contrast, the gcForest bypass architecture forms the layer input as

$$U^{(\ell)} := (X, Z^{(\ell-1)}),$$

which preserves the full Shannon information in X:

$$I\big(Y; U^{(\ell)}\big) = I(Y; X).$$

Therefore, feature concatenation does not create additional *Shannon* information about Y, but it guarantees that no label-relevant Shannon information carried by X is lost across layers.

Preservation without Improvement under Shannon Information.. Lemma 2 shows that the bypass concatenation mechanism in gcForest prevents the progressive loss of Shannon mutual information across layers. By forming the layer input as $U^{(\ell)} = (X, Z^{(\ell-1)})$, each layer preserves the full label-relevant information contained in the original input X, i.e.,

$$I\big(Y; U^{(\ell)}\big) = I(Y; X), \qquad \forall \ell.$$

This property eliminates the common issue of information degradation encountered in deep models that rely solely on successive transformations.

However, this preservation result also reveals a fundamental limitation of Shannon mutual information for explaining the behavior of deep forests. Since the maximal achievable mutual information between the representation and the label is already attained at the first-layer input $U^{(1)}$, the Shannon information perspective implies that subsequent layers cannot become strictly more informative about Y in an absolute sense. Under this view, deeper layers can at best preserve, but not exceed, the information available at earlier stages.

In practice, however, deep forests often exhibit improved predictive performance when additional layers are introduced, at least in certain regimes, while in other cases performance saturates after a small number of layers. Such behavior cannot be adequately explained within the classical information-theoretic framework, which treats all representations with identical Shannon mutual information as equally informative.

The root of this discrepancy lies in the implicit assumption underlying Shannon mutual information: it evaluates information with respect to an unrestricted, optimal decoder. In contrast, each layer of a deep forest employs a restricted class of learners—tree-ensemble predictors with finite depth, finite sample size, and limited representational capacity. As a result, although the total Shannon information about Y is preserved, different layers may expose different portions of this information to the learning algorithm.

This observation motivates the introduction of *usable information*, formalized as predictive V-information, which explicitly incorporates the capacity of a given predictive family. Within this framework, the central question becomes identifying the conditions under which deeper layers can increase usable information and thereby improve predictive performance.

2.5. Usable Information (Predictive V-information)

Following Xu et al. [1], we formalize "usable information" relative to a restricted family of predictors (models) that captures computational/statistical constraints.

Definition 2 (Predictive Family / Optional Ignorance [1])**.** Let $\Omega := \{f : \mathcal{X} \cup \{\emptyset\} \to \mathcal{P}(\mathcal{Y})\}$, where $f[x] \in \mathcal{P}(\mathcal{Y})$ denotes a predictive distribution for Y given side information x, and $f[\emptyset]$ denotes a predictive distribution without side information. A set $\mathcal{V} \subseteq \Omega$ is called a *predictive family* if it satisfies *optional ignorance*: for any $f \in \mathcal{V}$ and any $P \in \mathrm{range}(f)$, there exists $f' \in \mathcal{V}$ such that $f'[x] = P$ for all $x \in \mathcal{X}$ and $f'[\emptyset] = P$.

Definition 3 (Predictive Conditional V-Entropy [1])**.** Let (X, Y) be random variables on $\mathcal{X} \times \mathcal{Y}$ and let $\mathcal{V}$ be a predictive family. The *predictive conditional V-entropy* is

$$H_{\mathcal{V}}(Y \mid X) := \inf_{f \in \mathcal{V}} \ \mathbb{E}_{(x,y)\sim P_{X,Y}} \big[-\log f[x](y)\big],$$

and the *V-entropy* (no side information) is

$$H_{\mathcal{V}}(Y) := H_{\mathcal{V}}(Y \mid \emptyset) := \inf_{f \in \mathcal{V}} \ \mathbb{E}_{y\sim P_Y} \big[-\log f[\emptyset](y)\big].$$

Definition 4 (Predictive V-Information [1])**.** For a predictive family $\mathcal{V}$, the *predictive V-information from X to Y* is

$$I_{\mathcal{V}}(X \to Y) := H_{\mathcal{V}}(Y) - H_{\mathcal{V}}(Y \mid X).$$

Instantiation for gcForest (bypass).. Recall the bypass state (layer input) is

$$U^{(\ell)} := (X, Z^{(\ell-1)}), \qquad Z^{(\ell)} = f^{(\ell)}\big(U^{(\ell)}\big), \qquad U^{(\ell+1)} = (X, Z^{(\ell)}).$$

Given a (possibly layer-dependent) predictive family $\mathcal{V}^{(\ell)}$ representing the learner class used to exploit $U^{(\ell)}$ (e.g., bounded-complexity forests, or any restricted decoder family), we measure usable information at layer ℓ by $I_{\mathcal{V}^{(\ell)}}(U^{(\ell)} \to Y)$.

Proposition 3 (A Sufficient and Necessary Condition for Layer-wise Increase)**.** *Fix a predictive family $\mathcal{V}$ and consider a sequence of representations $U^{(1)}, U^{(2)}, \ldots$ with $U^{(\ell+1)} = T^{(\ell)}(U^{(\ell)})$ for some (possibly randomized) transformation $T^{(\ell)}$. Then the layer-wise usable information is nondecreasing,*

$$I_{\mathcal{V}}(U^{(\ell+1)} \to Y) \ \geq \ I_{\mathcal{V}}(U^{(\ell)} \to Y),$$

if and only if the predictive conditional V-entropy does not increase:

$$H_{\mathcal{V}}(Y \mid U^{(\ell+1)}) \ \leq \ H_{\mathcal{V}}(Y \mid U^{(\ell)}).$$

Moreover, the increase is strict *(i.e., $I_{\mathcal{V}}(U^{(\ell+1)} \to Y) > I_{\mathcal{V}}(U^{(\ell)} \to Y)$) if and only if*

$$\inf_{f \in \mathcal{V}} \mathbb{E}\big[-\log f[U^{(\ell+1)}](Y)\big] \ < \ \inf_{f \in \mathcal{V}} \mathbb{E}\big[-\log f[U^{(\ell)}](Y)\big] \, .$$

Proof. By Definition 4,

$$I_{\mathcal{V}}(U^{(\ell)} \to Y) = H_{\mathcal{V}}(Y) - H_{\mathcal{V}}(Y \mid U^{(\ell)}).$$

Since $H_{\mathcal{V}}(Y)$ is independent of $U^{(\ell)}$, comparing layers reduces to comparing $H_{\mathcal{V}}(Y \mid U^{(\ell)})$. The strict/non-strict statements follow immediately from the definition of $H_{\mathcal{V}}(Y \mid \cdot)$ as an infimum of expected log-loss over $\mathcal{V}$. □

Proposition 4 (When Deep Forest Layers Can Create Usable Information)**.** *Consider the gcForest bypass recursion $U^{(\ell+1)} = (X, f^{(\ell)}(U^{(\ell)}))$. Fix a predictive family $\mathcal{V}$ that represents the computationally bounded learner used downstream. If there exists a layer ℓ and a transformation $f^{(\ell)}$ such that the newly constructed feature $Z^{(\ell)} = f^{(\ell)}(U^{(\ell)})$ enables strictly better prediction of Y within $\mathcal{V}$, i.e.,*

$$\inf_{f \in \mathcal{V}} \mathbb{E}\big[-\log f[(X, Z^{(\ell)})](Y)\big] \ < \ \inf_{f \in \mathcal{V}} \mathbb{E}\big[-\log f[(X, Z^{(\ell-1)})](Y)\big] \, ,$$

then usable information strictly increases across that layer:

$$I_{\mathcal{V}}(U^{(\ell+1)} \to Y) \ > \ I_{\mathcal{V}}(U^{(\ell)} \to Y).$$

Conversely, if for some ℓ the best achievable log-loss within $\mathcal{V}$ saturates, i.e.,

$$\inf_{f\in\mathcal{V}} \mathbb{E}\big[-\log f[(X, Z^{(\ell)})](Y)\big] = \inf_{f\in\mathcal{V}} \mathbb{E}\big[-\log f[(X, Z^{(\ell-1)})](Y)\big]\,,$$

then $I_{\mathcal{V}}(U^{(\ell+1)} \to Y) = I_{\mathcal{V}}(U^{(\ell)} \to Y)$ and no further usable information is created at that layer for the family $\mathcal{V}$.

Interpretation ("when does depth help?").. Proposition 4 characterizes depth benefits in a way that is aligned with practice: usable information increases precisely when the newly constructed forest features $Z^{(\ell)}$ make the label more predictable *for the restricted learner family $\mathcal{V}$*. Thus, depth helps in regimes where a single-layer learner is capacity-limited (high approximation/estimation error), and additional layers produce features that reduce the achievable log-loss within $\mathcal{V}$; depth saturates when the restricted family can no longer exploit additional features to improve prediction.

Remark (Why this resolves the Shannon-information tension).. In the bypass architecture, Shannon mutual information with the label is preserved at every layer input: $I(Y; U^{(\ell)}) = I(Y; X)$ (cf. Lemma 2). Nevertheless, Propositions 3–4 show that $I_{\mathcal{V}}(U^{(\ell)} \to Y)$ can strictly increase with ℓ because the transformation $f^{(\ell)}$ can *re-express* information already present in X into forms that are more usable for $\mathcal{V}$, a phenomenon emphasized in Xu et al. [1] as "information can be created through computation."

Algorithm 1 Deep Forest

Input: xxx
Output: yyy

```
zzz
for t = 1 to T do
    xxx
end for
return yyy
```

References

[1] Yilun Xu, Shengjia Zhao, Jiaming Song, Russell Stewart, and Stefano Ermon. A theory of usable information under computational constraints. *arXiv preprint arXiv:2002.10689*, 2020.